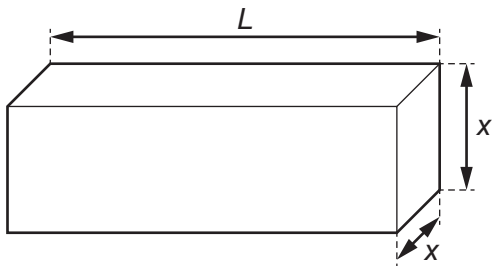


- 34** Conducting putty is a substance that conducts electricity and can be moulded into various shapes.

A cuboid made from this putty has length L and width x . The resistance between the square faces of the cuboid is $15\,\Omega$.



Some more of the putty is moulded into a cylinder that has length $\frac{L}{2}$.

The resistance between the circular faces of the cylinder is also $15\,\Omega$.

What is the radius of the cylinder?

A $\sqrt{\frac{x}{2\pi}}$

B $\frac{x}{\sqrt{2\pi}}$

C $\frac{x}{\sqrt{\pi}}$

D $\frac{\sqrt{2}x}{\sqrt{\pi}}$